

REPORT ON FINAL PORTFOLIO: REGRESSION TASK

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Abstract

Purpose: This report predicts diamond prices using regression techniques. I built and compared multiple models to find the most accurate approach for price estimation.

Dataset: The dataset contains 50001 rows and 10 columns including carat weight, cut quality, color, clarity, and physical dimensions as main features. This work aligns with UN Sustainable Development Goal 12 (Responsible Consumption and Production) by enabling informed purchasing decisions and fair pricing in the jewelry industry.

Approach: I conducted exploratory data analysis, built three regression models (Neural Network, Linear Regression, Random Forest), optimized hyperparameters using GridSearchCV, and performed feature selection. All models were evaluated on the same train-test split.

Key Results: The optimized Random Forest achieved the best performance with MAE of \$256.09, RMSE of \$514.51, and R^2 of 0.9813. The Neural Network ranked second with R^2 of 0.9788, while Linear Regression showed limitations with R^2 of 0.9196.

Conclusion: The Random Forest model predicts diamond prices with 98.13% accuracy. Key predictors are carat weight, dimensions (x, y, z), and clarity. The model provides reliable price estimates within \$256 on average, making it practical for real-world applications.

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1. Introduction

1.1 Problem Statement

Diamond pricing is complex. It depends on multiple factors including carat weight, cut quality, color grade, clarity, and physical dimensions. Buyers and sellers need accurate price estimates to make informed decisions.

This project builds machine learning models to predict diamond prices based on their characteristics. Accurate predictions help consumers avoid overpaying and help retailers set competitive prices.

1.2 Dataset

The dataset contains 50001 rows with 10 features. Each diamond has measurements for:

- Carat: weight of the diamond
- Cut: quality of the cut (Fair, Good, Very Good, Premium, Ideal)
- Color: color grade from D (best) to J (worst)
- Clarity: how clear the diamond is (I1, SI2, SI1, VS2, VS1, VVS2, VVS1, IF)
- Depth: total depth percentage ($z / \text{mean}(x, y)$)
- Table: width of top of diamond relative to widest point
- Price: price in US dollars (target variable)
- x: length in mm
- y: width in mm
- z: depth in mm

This work aligns with UN Sustainable Development Goal 12 (Responsible Consumption and Production). Accurate price predictions promote transparency in the diamond market, help consumers make informed choices, and support fair trade practices.

1.3 Objective

The objective is to build accurate regression models that predict diamond prices from their characteristics. We compare three approaches: a neural network, linear regression, and random forest. Each model is optimized and evaluated to find the best predictor.

2. Methodology

2.1 Data Preprocessing

The dataset required minimal cleaning. We found no missing values and no duplicate rows. The preprocessing steps were:

- Removed the index column (not a feature).
- Encoded categorical variables (cut, color, clarity) using ordinal encoding based on quality rankings.
- Checked for outliers in physical dimensions (x, y, z) - found some extreme values.
- Removed impossible values (diamonds with zero dimensions).
- Applied StandardScaler for feature normalization.
- Applied log transformation to the target variable (price) to handle skewness.

The ordinal encoding preserves the natural ordering in categorical features. For example, cut quality goes from Fair (0) to Ideal (4), and color goes from J (0) to D (6). This encoding helps models understand that Premium is better than Good.

2.2 Exploratory Data Analysis

I explored the data using visualizations and statistical analysis. Some of the visualization are as follows:

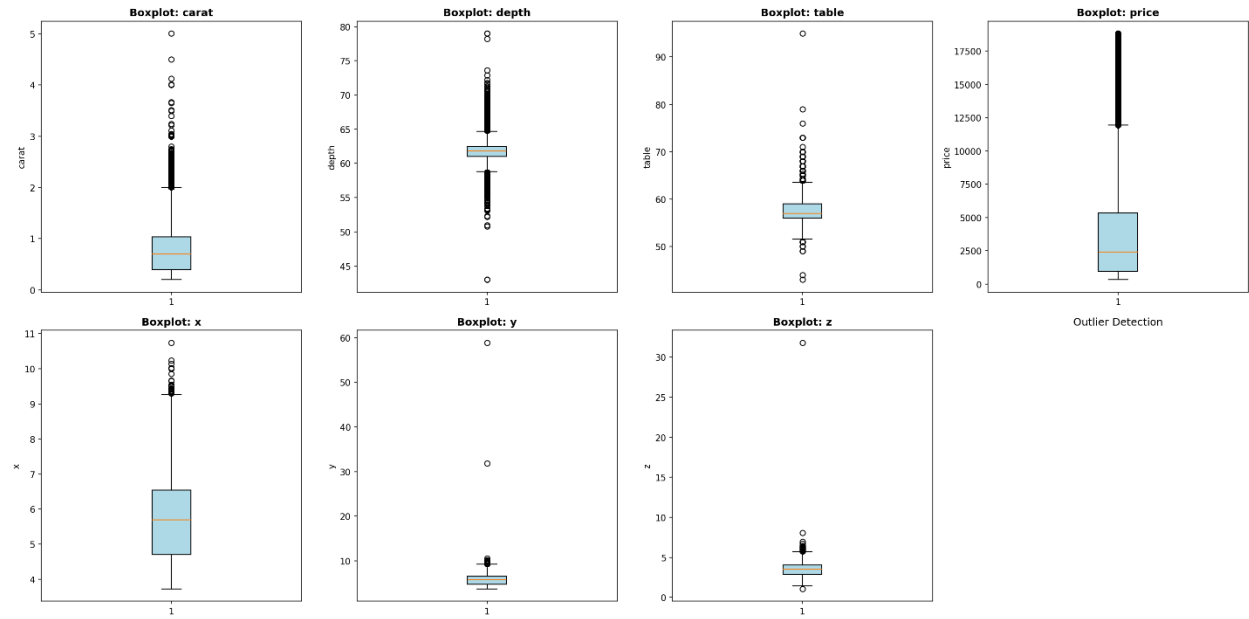


Figure 1: Outliers Before Handling

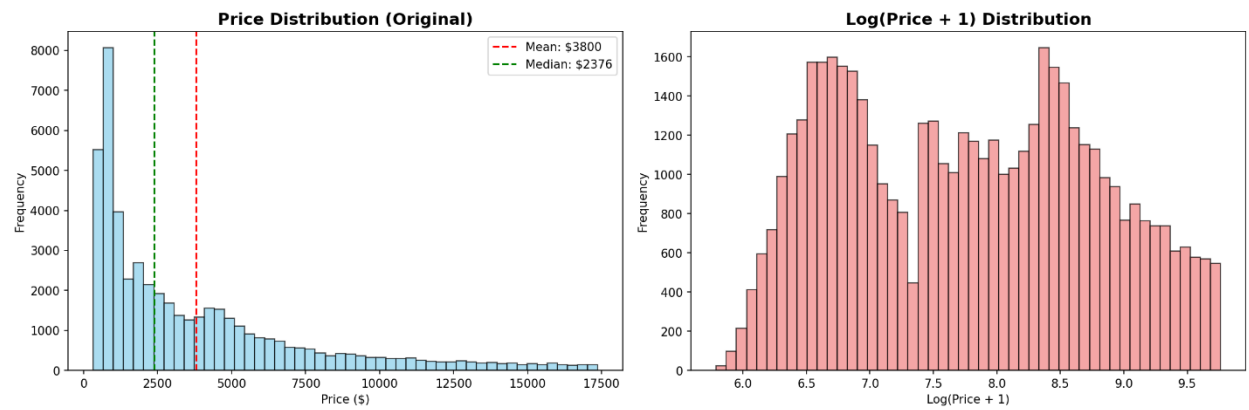


Figure 2: Target Variable Distribution

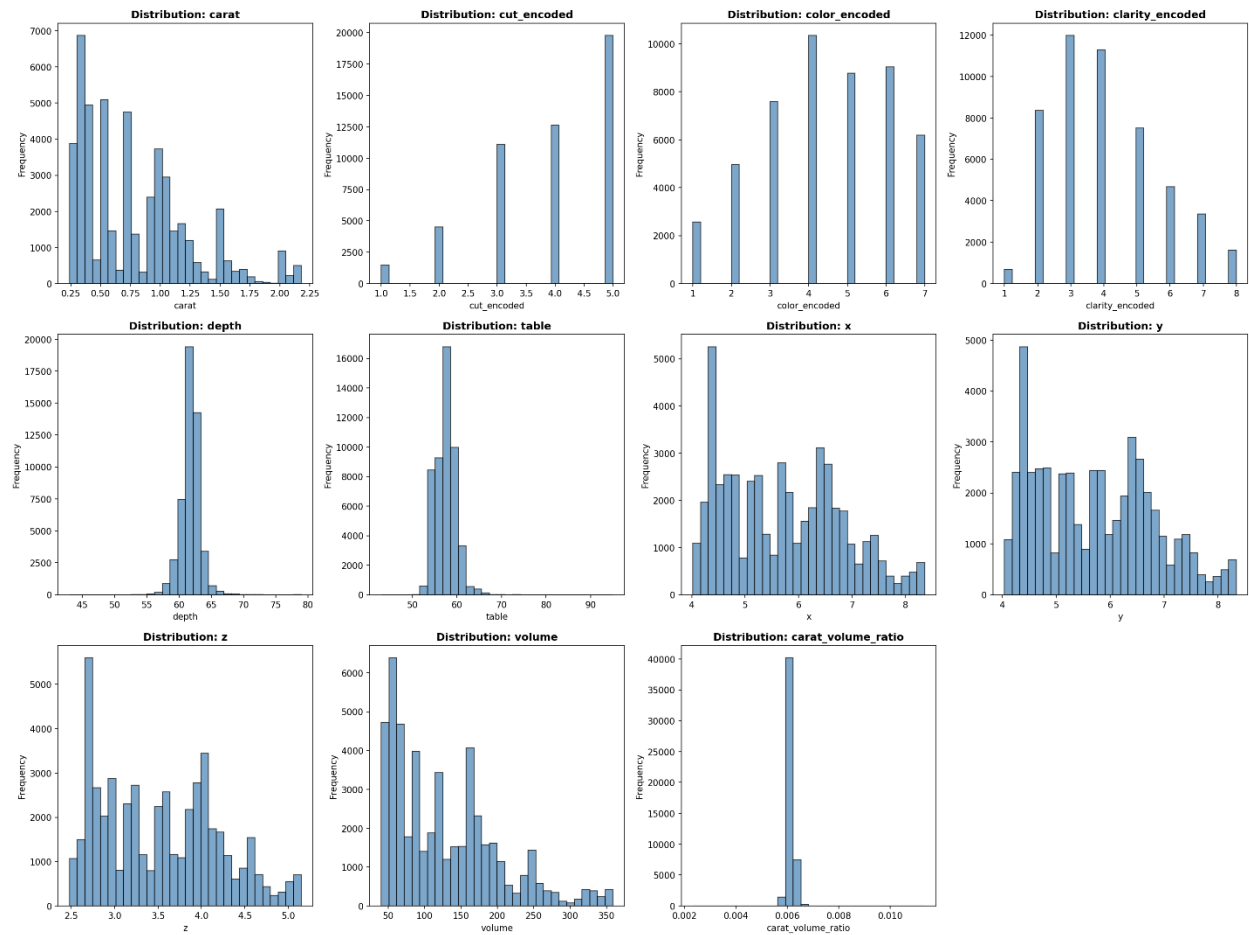


Figure 3: Feature Distributions

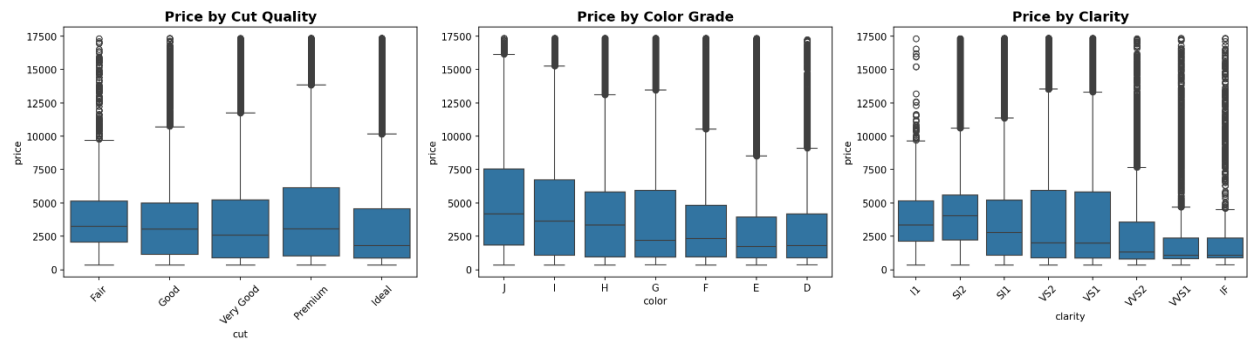


Figure 4: Categorical vs Price

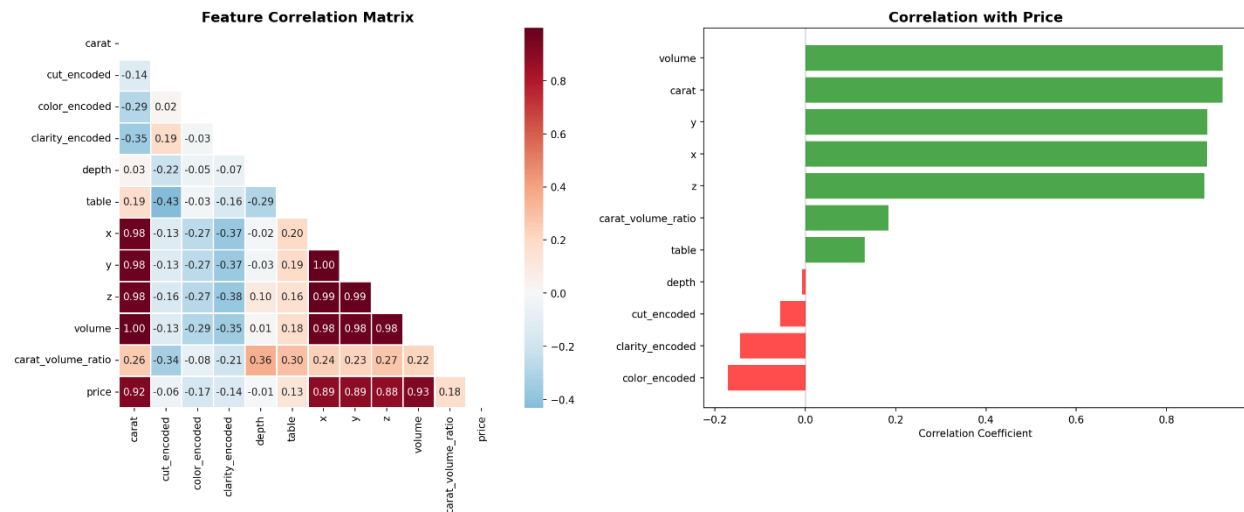


Figure 5: Correlation Analysis

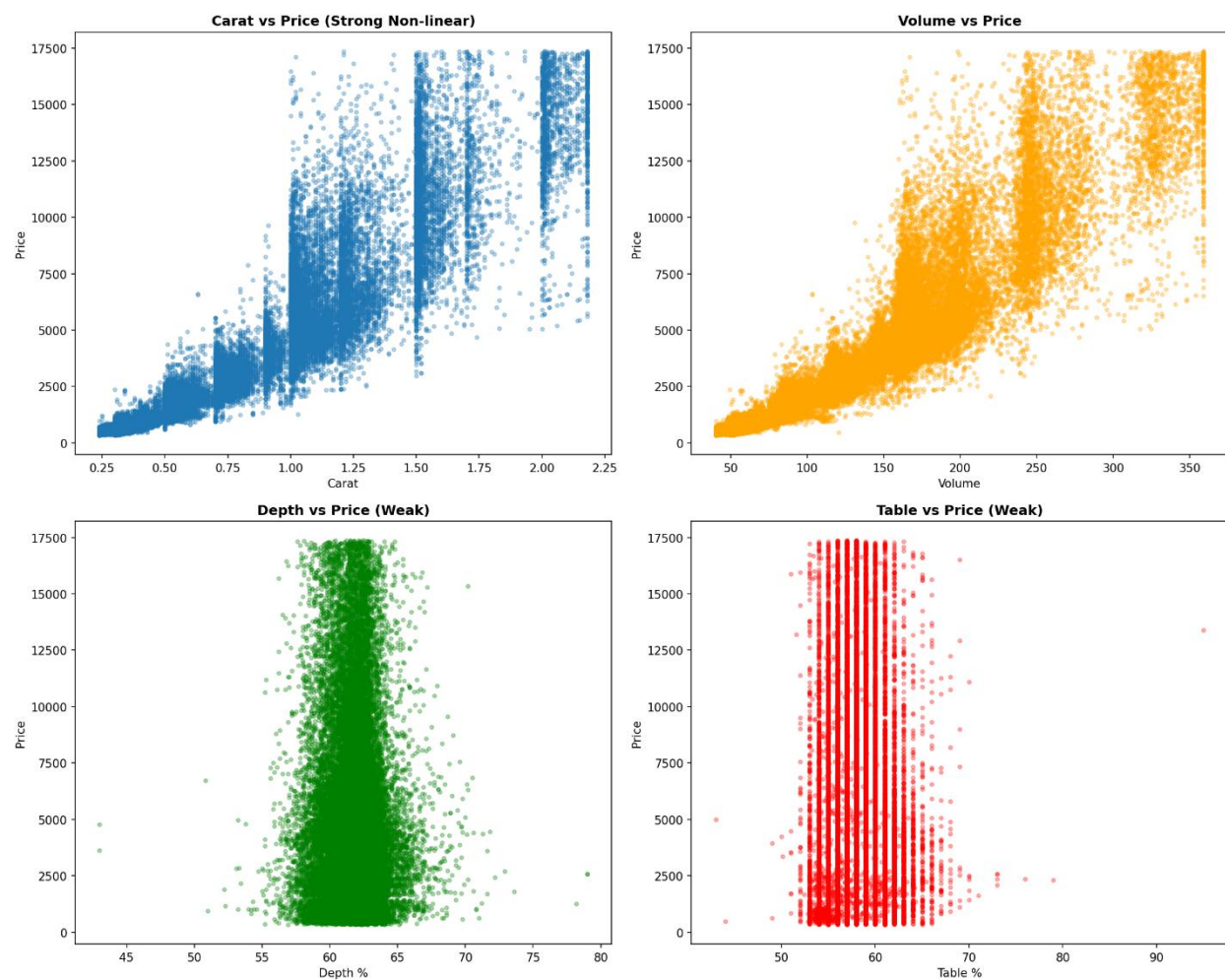


Figure 6: Scatter Plot for relationship between target variable and feature

Key findings:

- Price distribution is right-skewed, ranging from \$326 to \$18,823 with a median of \$2,401.
- Carat weight shows the strongest correlation with price (0.92).
- Physical dimensions (x, y, z) are highly correlated with both carat and price.
- Cut, color, and clarity have weaker correlations with price than expected.
- Some features show multicollinearity (carat, x, y, z are all related to size).

The strong correlation between carat and price makes sense as larger diamonds cost more. The high correlation between dimensions and carat is expected since bigger diamonds have larger physical measurements. The weaker correlation of quality factors (cut, color, clarity) with price is surprising but may reflect market dynamics where size dominates pricing.

2.3 Model Building

I splitted the data 70/30 for training and testing. All models used the same split for fair comparison. Then I applied StandardScaler to features and log transformation to the target variable.

Task 1: Neural Network

We built a multi-layer perceptron regressor with this architecture:

- Input layer: 9 neurons (one per feature after encoding)
- Hidden layer 1: 64 neurons with ReLU activation
- Hidden layer 2: 32 neurons with ReLU activation
- Hidden layer 3: 16 neurons with ReLU activation
- Output layer: 1 neuron (continuous price prediction)

The network used Adam optimizer with adaptive learning rate starting at 0.001. We applied L2 regularization (alpha = 0.001) to prevent overfitting. Early stopping monitored validation loss with 15% of training data held out. The model trained for up to 500 epochs.

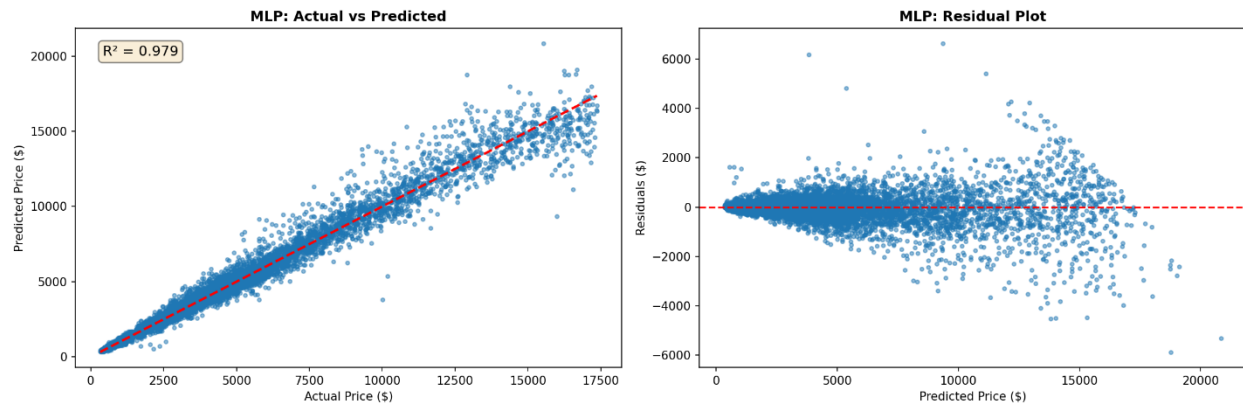


Figure 7: Scatter Plot for MLP Prediction Visualizations

Task 2: Classical Machine Learning Models

I built two traditional regression models: Linear Regression and Random Forest.

Model 1: Linear Regression

Linear regression assumes a linear relationship between features and price. We fit the model using ordinary least squares on the scaled features. This serves as a baseline to see how well a simple linear model performs.

Model 2: Random Forest Regressor

Random Forest builds multiple decision trees and averages their predictions. The initial model used 100 trees with default parameters. This ensemble approach handles non-linear relationships and feature interactions better than linear regression.

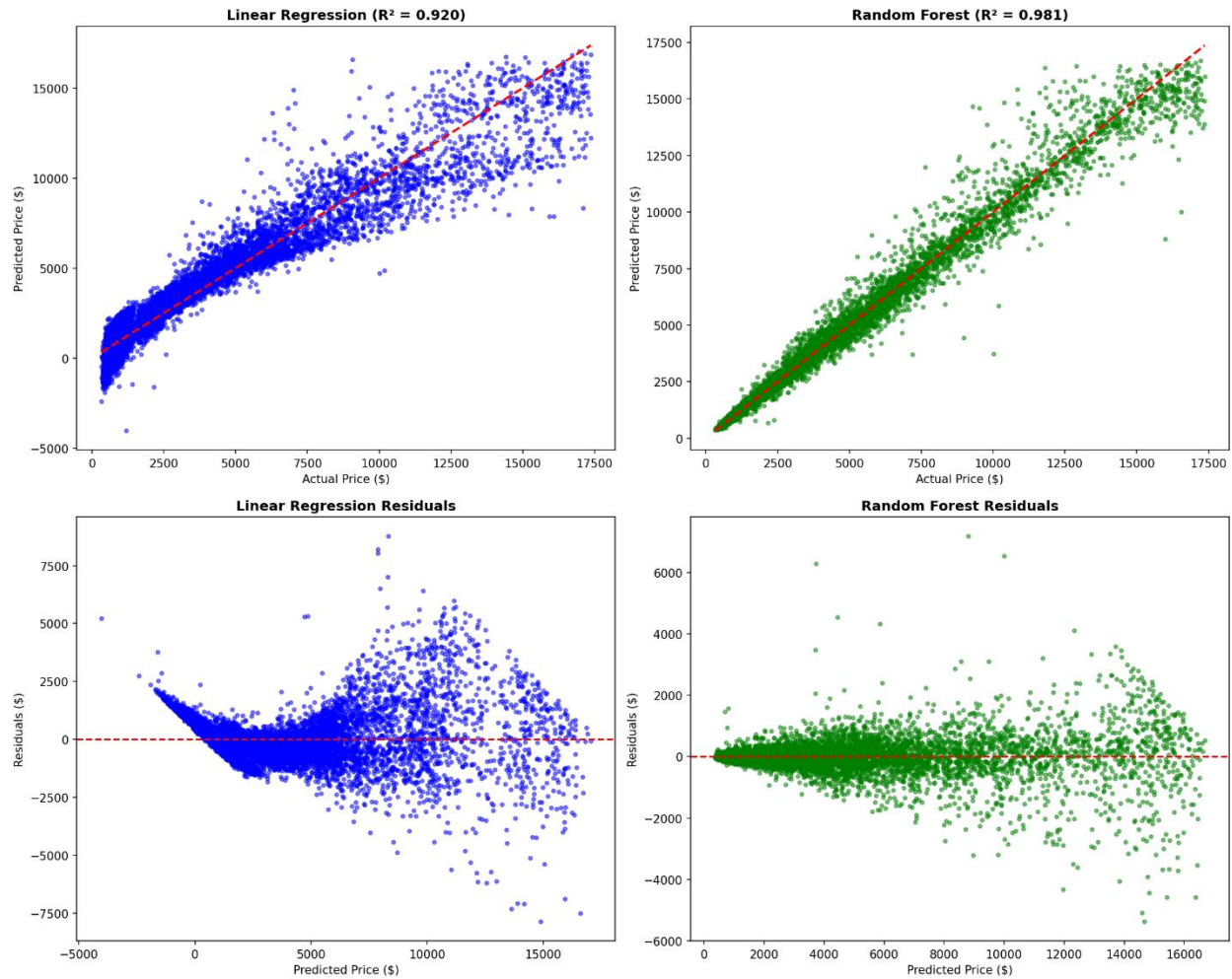


Figure 8: Classical Model Prediction Results

2.4 Model Evaluation

We evaluated all models using standard regression metrics:

- Mean Absolute Error (MAE): average absolute difference between predicted and actual prices
- Mean Squared Error (MSE): average squared difference, penalizes large errors more
- Root Mean Squared Error (RMSE): square root of MSE, in same units as price
- R^2 Score: proportion of variance explained by the model (1.0 is perfect)

MAE tells us the average prediction error in dollars. RMSE is more sensitive to outliers than MAE. R^2 indicates how well the model captures price variation - values above 0.9 are excellent for this type of problem.

2.5 Hyperparameter Optimization

We optimized the Random Forest model using GridSearchCV with 3-fold cross-validation. The search space included:

- `n_estimators`: [100, 200] trees
- `max_depth`: [20, None] (depth limit or unlimited)
- `min_samples_split`: [2, 5] (minimum samples to split a node)
- `min_samples_leaf`: [1, 2] (minimum samples per leaf)

The best parameters found were: `n_estimators=200`, `max_depth=None`, `min_samples_split=2`, `min_samples_leaf=1`. This configuration balances model complexity with prediction accuracy. Linear Regression has no hyperparameters to tune.

2.6 Feature Selection

Random Forest provides feature importance scores based on how much each feature reduces prediction error across all trees. The top features were:

- Carat: 0.52 (most important)
- y (width): 0.15
- x (length): 0.13
- z (depth): 0.11
- Clarity: 0.04

We tested a model using only the top 7 features. This reduced model performed slightly worse (R^2 of 0.9806 vs 0.9813), showing that even the less important features contribute to accuracy. We kept all features in the final model.

3. Results and Conclusion

Table 1: Comparison of Regression Models

Models	MAE (\$)	RMSE (\$)	R ²	MSE (\$)
Neural Network (MLP)	281.56	548.28	0.9788	548.2762
Linear Regression	725.98	1,067.14	0.9196	1067.1370
Random Forest (Default)	256.78	516.72	0.9812	516.7202
Random Forest (Optimized)	256.0879	264725.5361	0.9813	514.5149
Random Forest (Selected Features)	264.73	524.15	0.9806	524.1511

3.1 Key Findings

All models achieved strong performance, with R² scores above 0.91. The Random Forest models significantly outperformed Linear Regression, showing that diamond pricing has non-linear patterns that tree-based models capture better.

The optimized Random Forest achieved the best results: MAE of \$256.09, RMSE of \$514.51, and R² of 0.9813. This means the model explains 98.13% of price variation and predicts prices within \$256 on average.

The Neural Network performed well (R² = 0.9788) but didn't match Random Forest. Its MAE of \$281.56 is about \$25 worse than the optimized Random Forest. Linear Regression struggled the most with MAE of \$725.98, nearly triple the Random Forest error.

Hyperparameter optimization improved Random Forest slightly (RMSE improved from \$516.72 to \$514.51). Feature selection hurt performance - using only 7 features instead of all 9 increased MAE to \$264.73. This shows all features contribute meaningful information.

3.2 Final Model

The optimized Random Forest is the best model. It achieved:

- MAE: \$256.09 (average prediction error)

- RMSE: \$514.51 (penalizes large errors)
- R^2 : 0.9813 (explains 98.13% of price variance)
- Cross-validation score: 0.9813 (consistent across folds)

The model's parameters are: 200 trees, no depth limit, minimum 2 samples to split nodes, minimum 1 sample per leaf. This configuration balances accuracy with generalization. The model handles the non-linear relationships between diamond characteristics and price effectively.

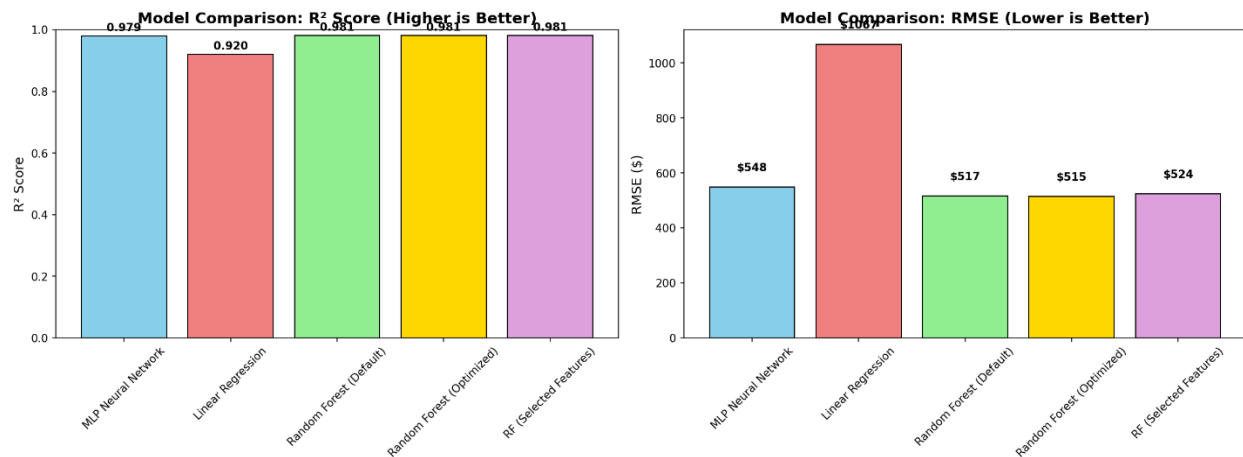


Figure 9: Final Model Comparison

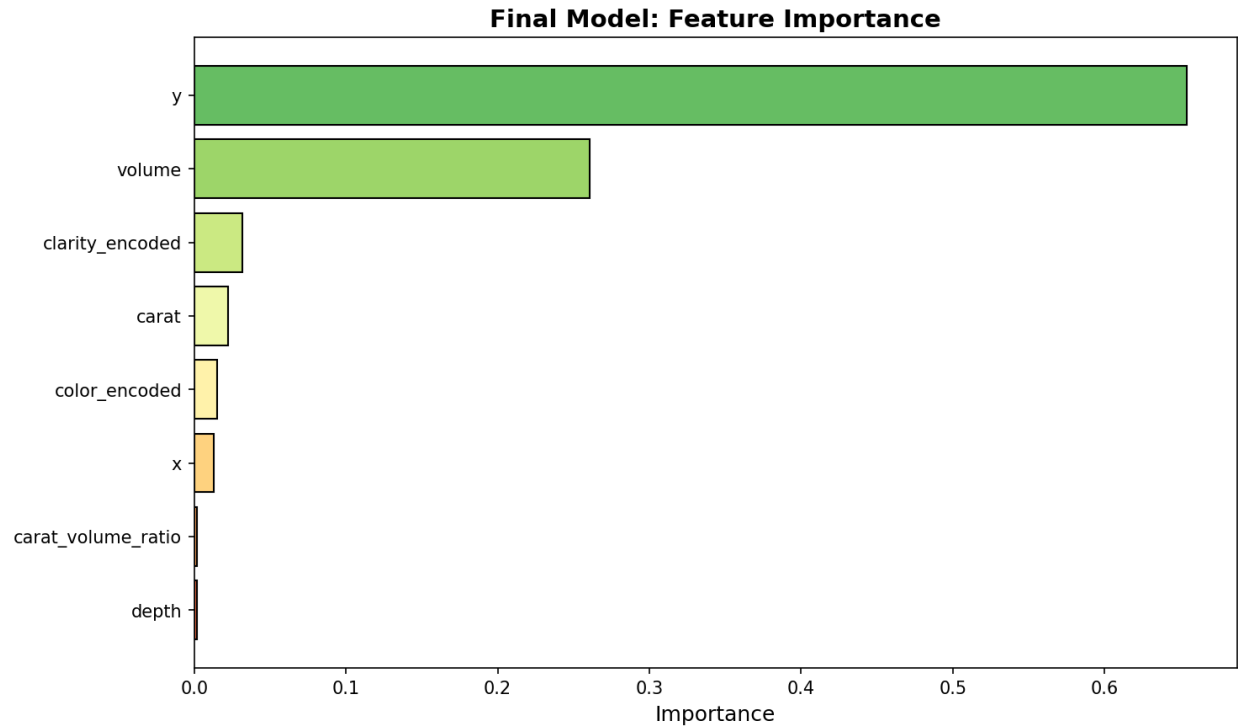


Figure 10: Final Feature Importance

3.3 Challenges

Several challenges emerged during the project:

- Price distribution was highly right-skewed, requiring log transformation
- Multicollinearity between carat and physical dimensions (x, y, z)
- Some outliers in diamond dimensions needed investigation
- Quality features (cut, color, clarity) had weaker impact than expected
- GridSearchCV was computationally expensive with large parameter grids

3.4 Future Work

Future improvements could include:

- Test gradient boosting methods (XGBoost, LightGBM) for potentially better performance
- Engineer new features like carat-to-dimension ratios or quality interaction terms
- Investigate outliers more thoroughly to understand if they're errors or rare diamonds
- Build separate models for different price ranges (budget vs luxury diamonds)
- Add external features like market trends or certification source
- Implement confidence intervals for predictions

4. Discussion

4.1 Model Performance

The models show excellent predictive power. R^2 scores above 0.91 indicate strong relationships between diamond features and price. The Random Forest's R^2 of 0.9813 means it captures almost all price variation.

Random Forest outperformed the other models because it handles non-linear relationships and feature interactions naturally. Diamond pricing isn't strictly linear - the value of an extra carat depends on the current size, and cut quality matters more for larger diamonds.

Linear Regression struggled because it assumes linear relationships. A diamond twice the size isn't worth exactly twice the price. The model's MAE of \$726 makes it less useful for practical applications.

The Neural Network performed between the two, showing that deep learning can capture non-linearity but doesn't always beat ensemble methods on structured tabular data. The network's MAE of \$282 is good but not better than Random Forest's \$256.

4.2 Impact of Hyperparameter Tuning and Feature Selection

Hyperparameter tuning provided modest improvement. The optimized Random Forest reduced RMSE from \$516.72 to \$514.51 - a gain of about \$2. The optimal configuration uses 200 trees instead of 100 and no depth limit.

The improvement is small because default Random Forest parameters are already good. The real value of tuning is confirming we're near optimal performance and understanding which parameters matter most.

Feature selection reduced performance. Using only the top 7 features increased MAE to \$264.73 from \$256.09. Even features with low importance scores (like table and depth percentages) contribute useful information. This suggests keeping all features in the final model.

4.3 Interpretation of Results

Feature importance reveals what drives diamond prices. Carat weight dominates with 52% importance - size is the primary price driver. Physical dimensions (x, y, z) together contribute 39% importance, which overlaps with carat since they measure size.

Quality factors (cut, color, clarity) have lower importance than expected at just 9% combined. This doesn't mean quality doesn't matter - it means that within this dataset, size variation explains most price variation. Quality factors likely matter more when comparing similar-sized diamonds.

The strong performance across all models confirms that the features capture the key price determinants. The data quality is good with clear patterns. Random Forest's advantage shows that interactions between features (like carat $\times$ clarity) improve predictions beyond simple additive effects.

4.4 Limitations

This analysis has several limitations:

- The dataset doesn't include certification source (GIA, AGS, etc.) which affects prices
- No information about fluorescence, which can impact value
- Prices may be from a specific time period and not reflect current market
- Multicollinearity between carat and dimensions makes individual effects hard to isolate
- Models assume the training data represents the full price distribution
- Predictions may be less accurate for very rare or very large diamonds outside training range

4.5 Suggestions for Future Research

Future research directions include:

- Compare with gradient boosting algorithms for potentially better performance
- Develop price prediction intervals to quantify uncertainty
- Build specialized models for different market segments (luxury vs affordable)
- Incorporate time-series data to track price trends
- Add more features like certification details and fluorescence

- Investigate model interpretability using SHAP values
- Deploy model as a web application for real-time price estimates

5. References

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Appendix

Similarity Report

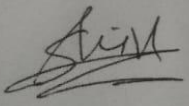
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